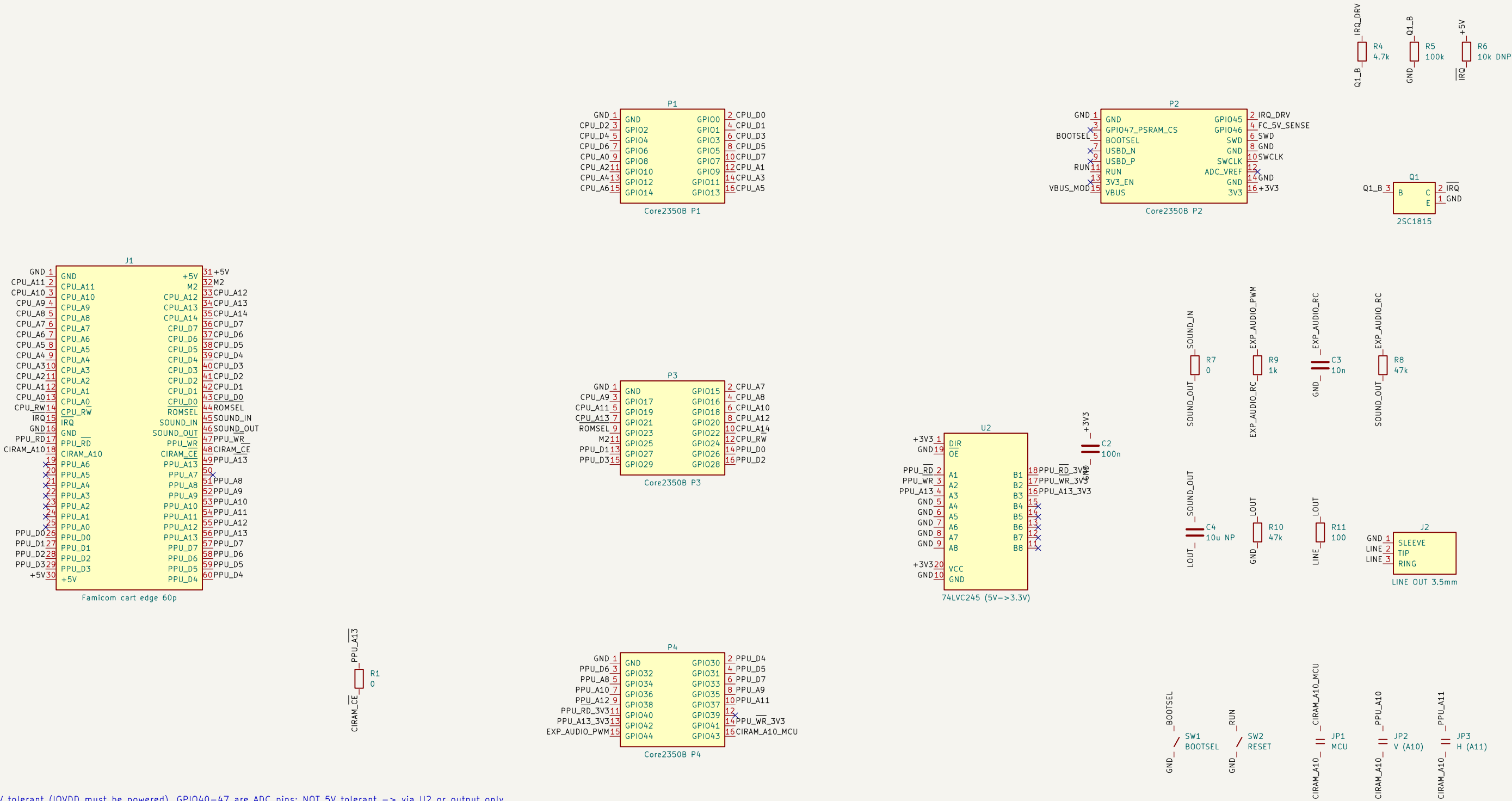


FC-MAGICON: Waveshare Core2350B (RP2350B) Famicom cartridge – PRG/CHR emulation, NSF, screen streaming



GPIO0–38 are 5V tolerant (IOVDD must be powered). GPIO40–47 are ADC pins: NOT 5V tolerant → via U2 or output only.

PPU A0–A7 (cart 19–25, 50) are NOT used: low address is sampled on PPU_D0–7 (=PPU AD0–7) just before /RD falls.

GPIO39 has the on-board LED (470R) – keep it off the bus. GPIO47 = on-board PSRAM CS.

RP2350 erratum E9: do not enable internal pull-downs on bus pins.

Close exactly one of JP1/JP2/JP3 (CIRAM A10 source).

Firmware must keep all bus pins as inputs until FC_5V_SENSE is high (USB-only power must not back-power the console).

Cart pins 1,16,30,31,15,18,45,46,47,48,49: nesdev docs, cross-checked with fc-rom-vomitter (51 signal pins verified on real carts).

Q1 25C1815 pin order E–C–B (flat face). R6 is DNP unless the console has no /IRQ pull-up.

R8 (mix level) and R7 (pass-through) values are to be tuned on real hardware.

Core2350B header pinout: Waveshare schematic, checked against module silk on 2026–09–26. Module USB uses its own FPC adapter.